9 **(a)** In a circuit, a 4.0 V battery with internal resistance is connected in series to a thermistor and a coaxial resistor.

The coaxial resistor has a cylindrical core of length 0.020 m. The core is made of material A and is surrounded by material B as shown in Fig. 9.1. In the circuit, the direction of the current passing through the coaxial resistor is parallel to its length. The physical dimensions of the coaxial resistor are fixed.

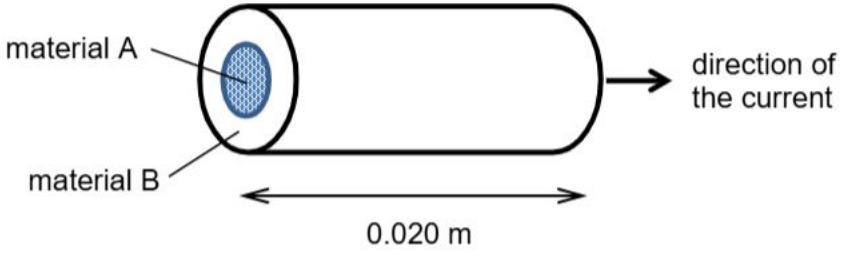


Fig. 9.1

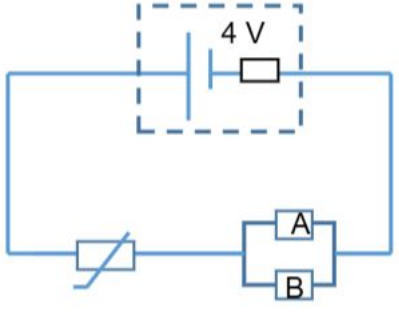
(i) Draw a complete circuit diagram of the circuit in the space below.

Represent the coaxial resistor as two separate resistors A and B, one for each material respectively.

thermistor B1

parallel B1

internal resistance B1



[3]

(ii) Fig. 9.2 shows the variation of the current I in the thermistor with temperature T of the thermistor.

Fig 9.3 shows the variation of I in the thermistor with voltage V across the thermistor.

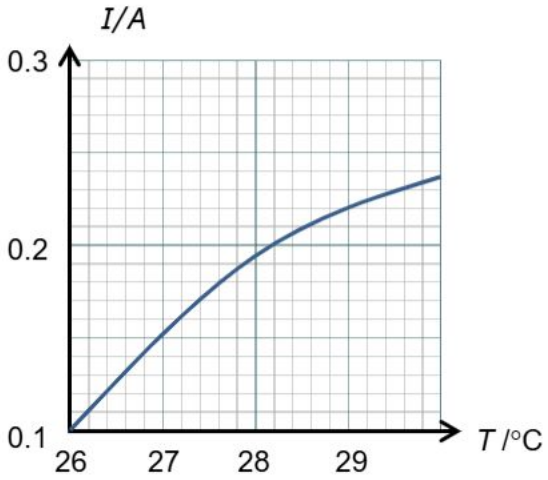


Fig. 9.2

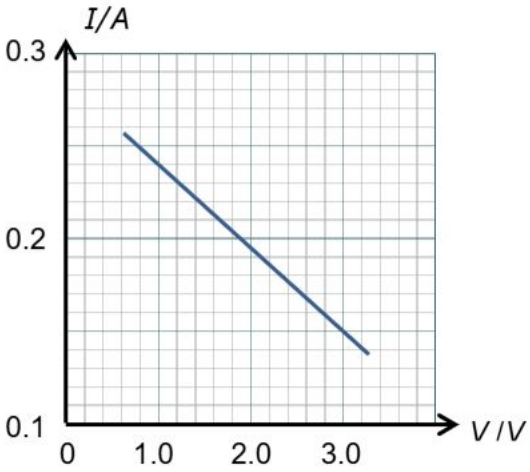


Fig. 9.3

Determine the resistance of the thermistor at 27 °C.

Reading from the graph, at 27°C, I = 0.15 A

And pd across resistor = 3 V C1

Hence resistance of thermistor = $\frac{3}{0.15} = 20\ \Omega$ A1

resistance = ... Ω [2]

(iii) Describe and explain what will happen to the potential difference across resistors A and B in the circuit when the temperature of the thermistor increases.

By potential divider rule, $V_A = \frac{R_{effA,B}}{\text{Total resistance}} \times E$ M1

Since total resistance drop, V_A and hence V_B increases since they are in parallel. A1

.....[2]

(b) (i) Two identical lamps are connected to a d.c. and a.c. supply as shown in Fig. 9.4a and Fig. 9.4b respectively.

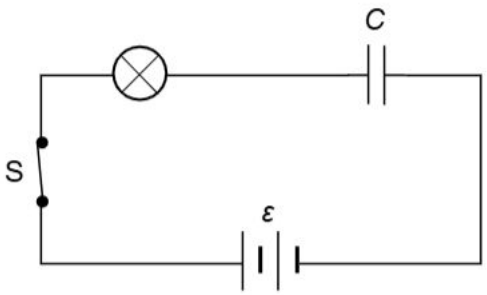


Fig 9.4a

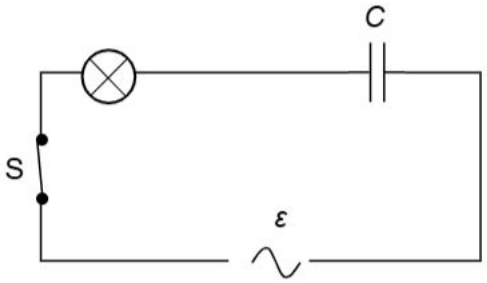


Fig 9.4b

State and explain which lamp will be turned on for a longer period of time. Assume that the time constant is large.

In 9.4a, the lamp will be on when the capacitor is charging. At full charge, the lamp will be off B1

In Fig 9.4b, when the capacitor is constantly charged and discharged because of the periodic nature of the a.c.

Hence a continuous supply of ac passes through the lamp to light it and it will stay lit for a longer period of time.

[3]

(ii) The charged capacitor of capacitance C is being discharged through a resistor of resistance R, as shown in Fig. 9.5. At time = 0 s, the capacitor is fully charged.

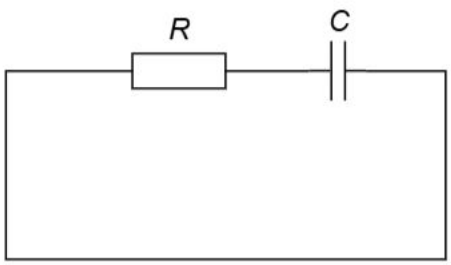


Fig. 9.5

The energy stored in the capacitor decreases with time as the capacitor discharges. Determine the time t in terms of R and C, it takes for this stored energy to be discharged by 75% of its initial value.

Energy = $\frac{q^2}{2C}$

$= \frac{Q^2}{2C} e^{-\frac{2t}{RC}}$ B1 (if took as 3/4 can get method mark for this)

$= U_0 e^{-\frac{2t}{RC}}$

Hence $\frac{1}{4} U_o = U_0 e^{-\frac{2t}{RC}}$ M1

$t = \frac{1}{2} RC \ln 4 = 0.693RC$ A1

t = ... [3]

(c) After a long time, the capacitor in Fig. 9.5 is fully discharged and a battery of e.m.f E of negligible internal resistance is used to recharge the capacitor in a circuit as shown in Fig. 9.6.

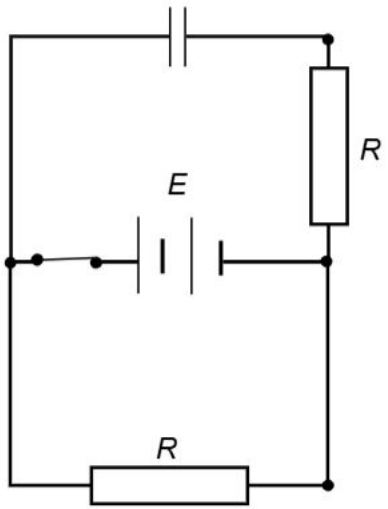


Fig. 9.6

(i) Determine the current, in terms of E and R, in the battery immediately after the switch is closed.

$R_{eff} = \frac{R}{2}$ M1

$I = \frac{E}{2R}$ A1

current = ... [2]

(ii) Determine the current, in terms of E and R, in the battery a long time after the switch is closed.

When the capacitor is fully charged, there is no current in the resistor in series with it. M1

Hence $I = \frac{E}{R}$ A1

current = ... [2]

(iii) Hence suggest an expression for the time constant, in terms of R and C.

Explain your reasoning.

The capacitor and its resistor (the one in series with it) are parallel to the lower resistor.

B1 Hence, the pd across the lower resistor is always E and thus (will not affect) can be removed without affecting the charging process. M1 Hence, only the resistor in series with it will affect the charging process and thus the time constant is RC. A1

[3]

[Total: 20]